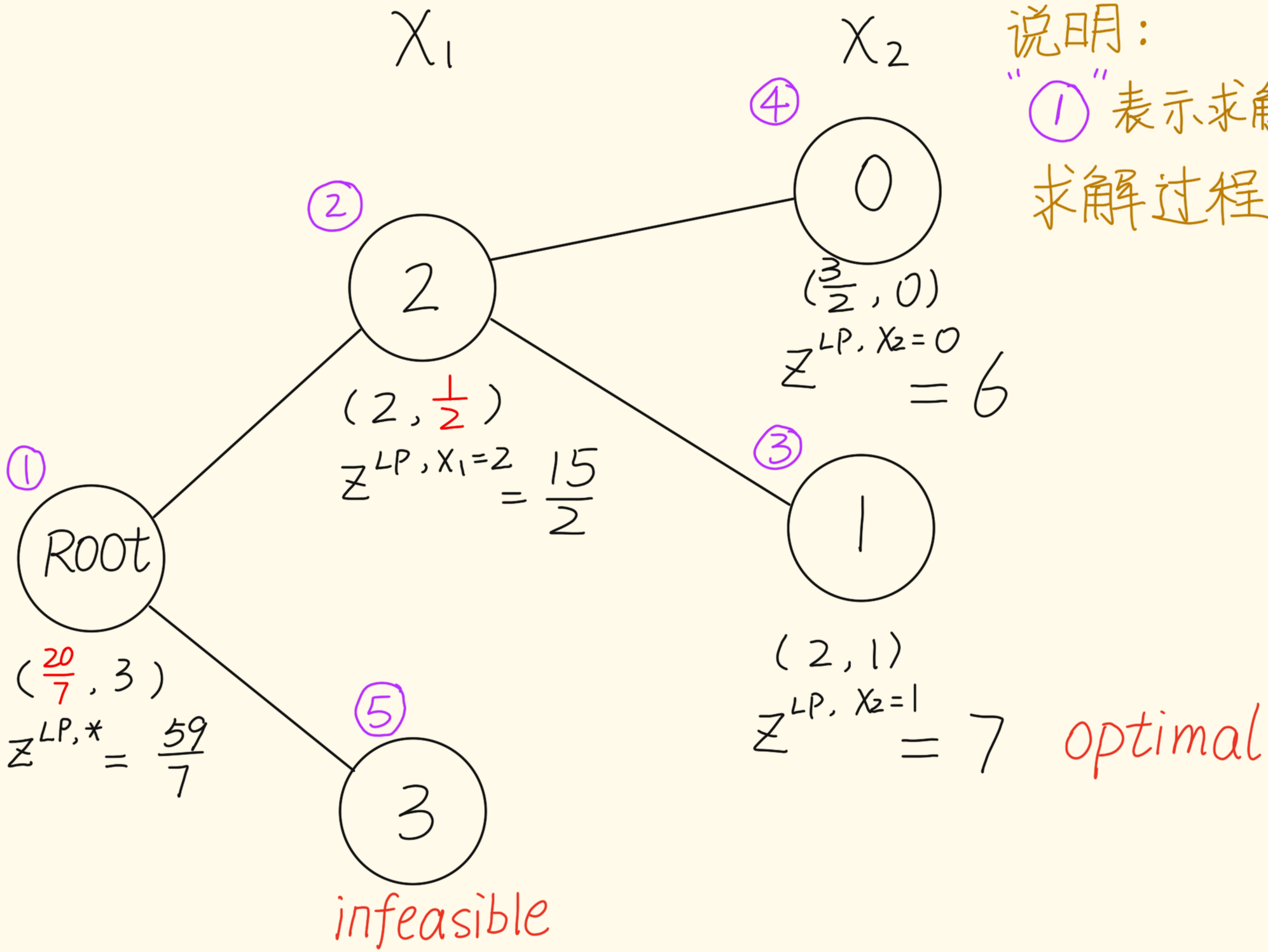

说明：

"①"表示求解顺序

求解过程在Excel中

Step 1: Solve relaxed LP Model

$$\max \quad z = 4x_1 - x_2$$

$$\text{s.t.} \quad 7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

$$x_1, \quad x_2 \geq 0$$

feasible region separation : constraints in LP Model

the optimal solution : $(x_1, x_2)^T = (\frac{20}{7}, 3)$

$$z = \frac{59}{7}$$

$UB = \frac{59}{7}$ $\quad$ $LB = -inf$

incumbent solution : None

x_1 is non-integer, so add two more nodes

Node 2 : $x_1 = 2$

Node 3 : $x_1 = 3$

Step 2: Add constraint $x_1 = 2$

$$\max \quad z = 4x_1 - x_2$$

$$\text{s.t.} \quad 7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

$$x_1, \quad x_2 \geq 0$$

feasible region separation:

constraints in LP Model and $x_1 = 2$

the optimal solution: $(x_1, x_2)^T = (2, \frac{1}{2})$

$$z = \frac{15}{2}$$

$UB = \frac{15}{2}$ $\quad$ $LB = -\inf$

incumbent solution: None

x_2 is non-integer, so add two more nodes

Node 4: $x_2 = 0$

Node 5: $x_2 = 1$

Step 3 : Add constraint $x_2 = 1$

$$\max \quad z = 4x_1 - x_2$$

$$\text{s.t.} \quad 7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

$$x_1, \ x_2 \geq 0$$

feasible region separation :

constraints in LP Model and $x_2 = 1$

the optimal solution : $(x_1, x_2)^T = (2, 1)$

$$z = 7$$

$UB = \frac{15}{2}$ $\quad LB = 7$

incumbent solution : 7

Both x_1 and x_2 is integer , so update incumbent solution and lower bound (LB)

Step 4: Add constraint $x_2 = 0$

$$\max \quad z = 4x_1 - x_2$$

$$\text{s.t.} \quad 7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

$$x_1, \quad x_2 \geq 0$$

feasible region separation:

constraints in LP Model and $x_2 = 0$

the optimal solution: $(x_1, x_2)^T = (\frac{3}{2}, 0)$

$z = 6$

$UB = \frac{15}{2}$ $\quad LB = 7$

incumbent solution: 7

Step 5: Add constraint $x_1 = 3$

$$\max \quad z = 4x_1 - x_2$$

$$\text{s.t.} \quad 7x_1 - 2x_2 \leq 14$$

$$x_2 \leq 3$$

$$2x_1 - 2x_2 \leq 3$$

$$x_1, \; x_2 \geq 0$$

feasible region separation:

constraints in LP Model and $x_1 = 3$

the optimal solution: infeasible

$UB = \frac{15}{2}$ $\quad LB = 7$

incumbent solution: 7

the optimal solution to the IP Model

is $(x_1, x_2)^T = (2, 1)$

$$\text{Max } z = 4x_1 - x_2 = 7$$